



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SHRI VALLAPA GANAPATHY AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

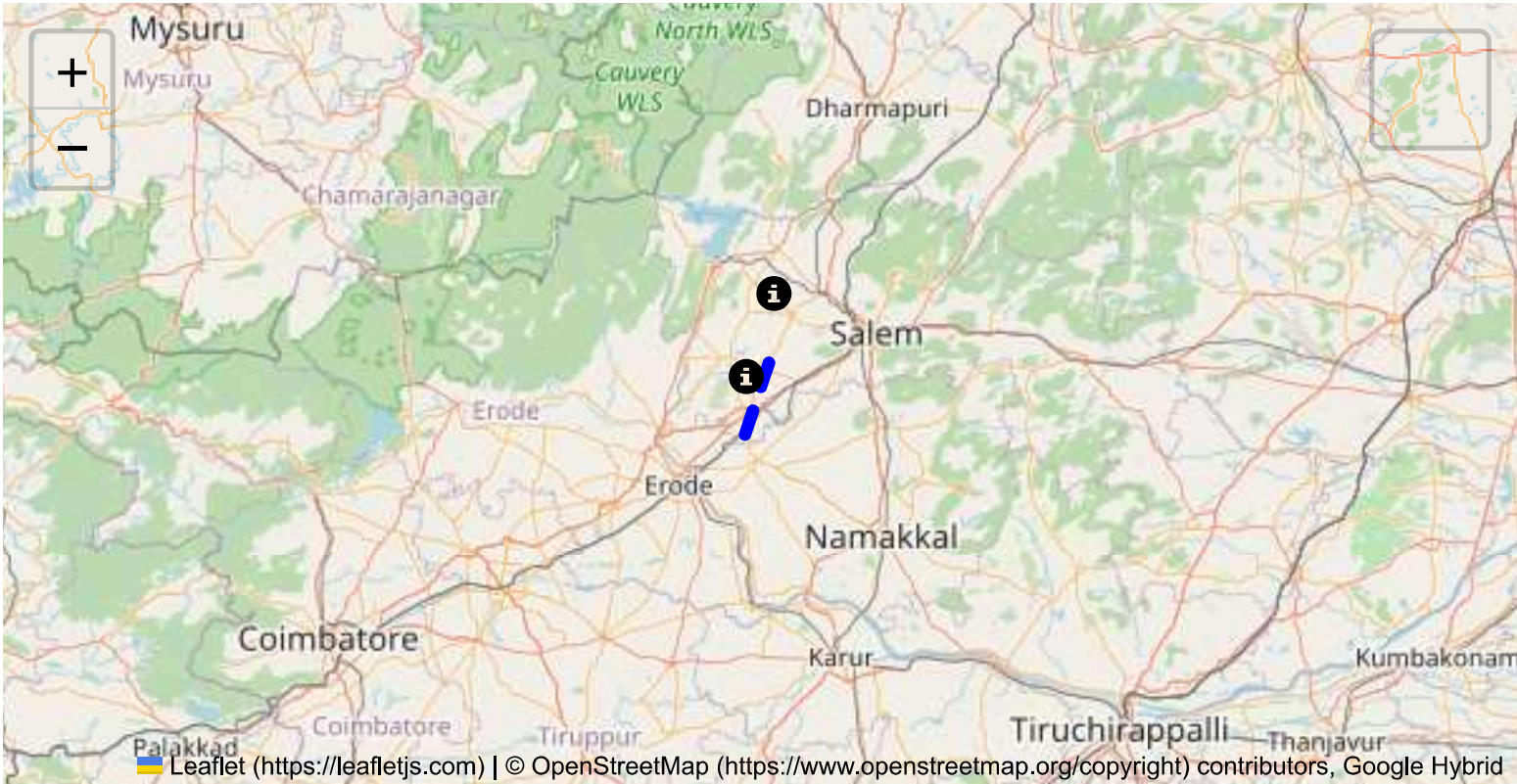
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

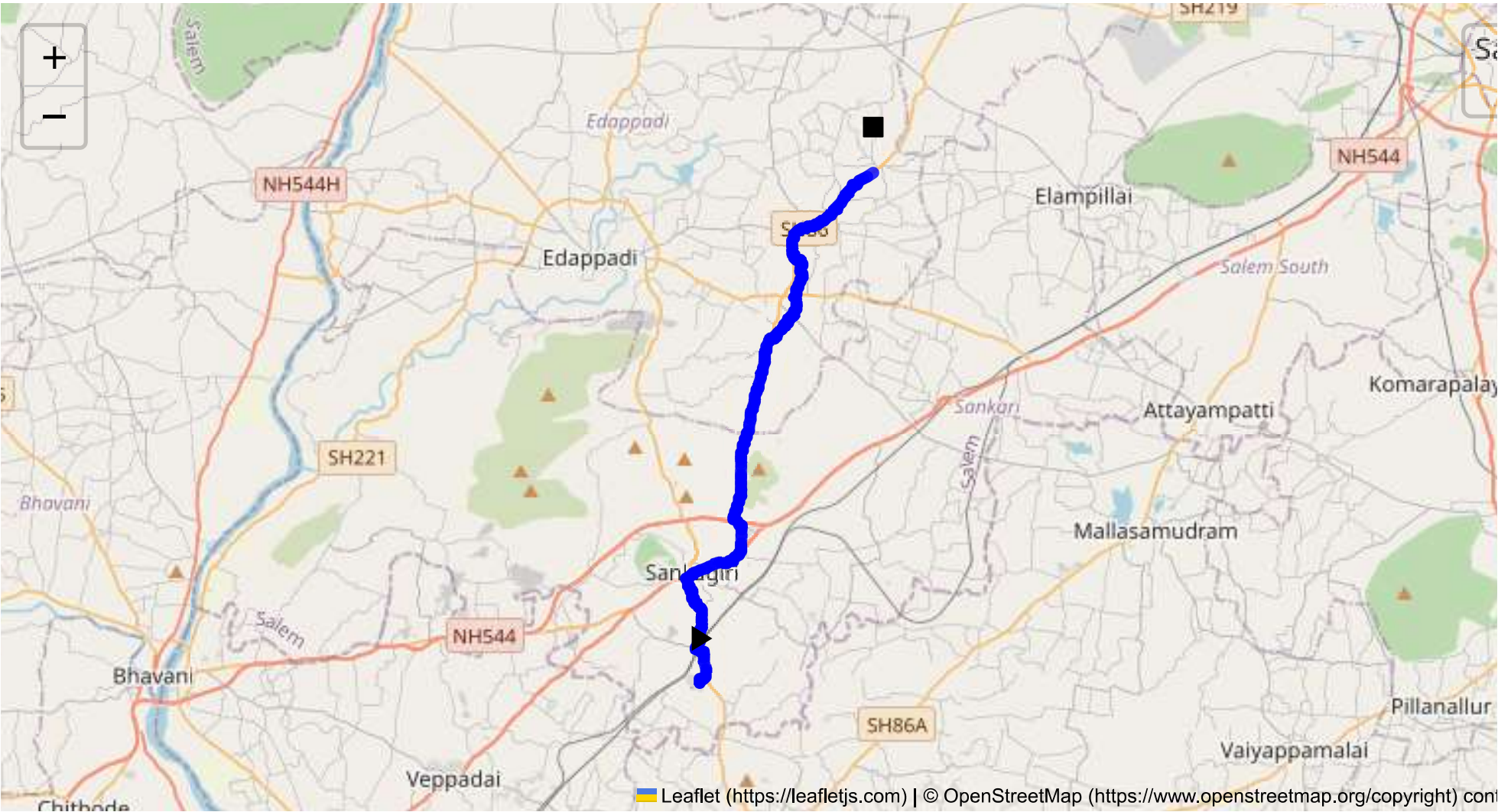
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 23.81 km
Estimated Duration: 0.6 hours
Adjusted Duration (Heavy Vehicle): 0.8 hours
Start: (11.4381, 77.8734)
End: (11.61301051, 77.93308361)



Welcome to the Journey Risk Management Study

Route Analysis: CVQF+23W, Sangagiri to JW7M+476, Koranampatti

1. Overview of the Route Map:

- The route covers approximately 23.81 kilometers, primarily along local roads with a combination of rural and semi-urban settings. The starting point is in Sangagiri, and the destination is

Koranampatti, passing through the NH44 and SH86 highways. The travel time for heavy vehicles is estimated at about 38 minutes.

2. Typical Weather Conditions and Hazards:

- Tamil Nadu has a tropical climate with hot and humid weather. The monsoon from October to December can bring heavy rain, leading to potential flooding and waterlogging, particularly on poorly drained roads. During summer, high temperatures may impact vehicle performance.

3. Traffic Patterns:

- Peak traffic hours are typically in the morning (8-10 AM) and late afternoon (5-7 PM). Congestion is more likely near market areas and highway intersections, such as near NH44 junctions. Weekend traffic might increase as more local travel occurs.

4. Assessment of Road Quality and Infrastructure:

- Major highways like NH44 and SH86 generally have good road quality with regular maintenance, but rural roads may have uneven surfaces and potholes. Watch for narrow sections and areas with construction that may lack proper signage.

5. Alternative Routes for Emergencies:

- An alternative is via NH544 ensuring you maintain travel on larger, well-maintained roads. This may increase travel distance but offers better infrastructure and emergency access.

6. Local Regulations for Hazardous Material Transport:

- Transporting hazardous materials requires adhering to specific routes designated by local authorities, a valid license for transport, and appropriate signage on the vehicle. Night transport might be restricted for safety purposes.

7. Historical Incidents:

- There have been events involving vehicle rollovers and minor spills in narrow rural areas due to sharp turns. Previous incidents emphasize cautious driving, particularly under adverse conditions or on unfamiliar routes.

8. Environmental Considerations:

- Sensitive areas include agricultural lands and small water bodies. Measures should be in place to prevent contamination in case of spills. Wildlife crossings may be present; drivers should watch for animal signs.

9. Communication Coverage:

- Overall, good cellular coverage is expected, but potential dead zones might exist in rural stretches between villages. Drivers should ensure their cellular devices are fully charged and carry alternative communication tools like CB radios.

10. Emergency Response Times:

- In populated areas near highways, emergency response times can range from 15 to 30 minutes. In rural or less accessible areas, response times might extend to an hour or more. Knowing local police stations and hospitals along the route is advisable.

11. Overall Summary of Risk Assessment:

- The route is moderately safe with proper preparedness. Key risks include weather-related issues during monsoon, peak traffic congestion, and environmental hazards. Regular checks on vehicle condition and emergency protocols, along with staying informed of local regulations, are vital for reducing risks. Emphasize defensive driving and constant communication with the logistics team to manage unexpected events efficiently.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	Medium	11.43971, 77.87348	30 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.45007, 77.87201	15 KM/Hr
5	Turn	High	11.47407, 77.86839	15 KM/Hr
6	Turn	Medium	11.48363, 77.88780	30 KM/Hr
7	Turn	Medium	11.48446, 77.88787	30 KM/Hr
8	Turn	Medium	11.49451, 77.88489	30 KM/Hr
9	Turn	High	11.55614, 77.89764	15 KM/Hr
10	Turn	High	11.58317, 77.90647	15 KM/Hr
11	Turn	Medium	11.59061, 77.90583	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
1	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium